Time series analysis and forecasting of Austrian electricity prices using ARIMA/ARIMAX modeling

Capstone Project for the 2025 „Data Practictioner“ Class at neuefische

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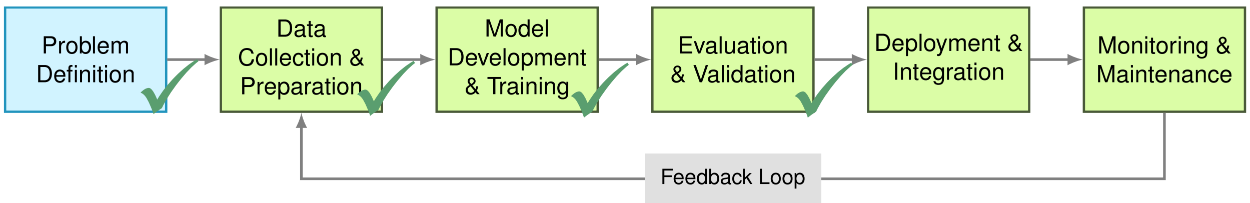
Business Case: Power Consumers

- Following the russian full-scale invasion of Ukraine in 02/2022, Austrian energy prices experienced significant increases and have also become very volatile.
- Large power producers (e.g. wind/solar parks, public utility companies) and power consumers (e.g. manufacturing companies, data centers) need planning security.
- This is best achieved by securing over-the-counter supply contracts. These are sold at fixed prices for different durations (e.g. 1-6 months-ahead).
- Forecasting these prices yields benefits in terms of risk-management as well as cost optimization in procurement planning for power consumers.
- This project explores the cornerstones of a light-weight, continuous forecasting application designed for use by small-scale manufacturing companies in Austria.

Project Scope

- **Data sources**: Current real-world sources only! Historical data as stand-in for daily-updated market data
- **Data Engineering**: Development of a semi-automatic ETL-pipeline prototype
- **Data Preparation & Analysis**: EDA including time-series decomposition + basic Feature Engineering
- **Model building**: Design and Evaluation of prognosis model (ARIMAX & Regression).
- **Tableau Story for Visualisation**
- **Architecture, analytics, methodology, model building decisions by author.** Code developed using AI-assisted pair programming (AI-generated, manually validated and documented).



Source: https://www.mlsysbook.ai/contents/core/workflow/workflow_files/mediabag/9b58b3aa72de00d638019e0cdf45f488ecf3911f.svg

Data Overview

Data Granularity: Monthly

- **Outcome Variable:** Average of day-ahead electricity prices (EXAA / Vienna Stock Exchange)
- **Input Variables**:
    - Gas prices - Average of 1-month ahead futures (Austrian Energy Agency)
    - Price of carbon emission credits in European Trading Area (ICAP)
    - Climate data for austria (EUROSTAT)
    - Proportion of renewables in energy production (E-Control)
    - Economic Indicators (STATISTIK Austria)

Central EDA Finding

Three distinct periods in electicity price data:

Period	~ Timespan	No. of months	Mean (EUR/MWh)	SD	Stationary?
Pre-Shock	2015-2020	72	35,94	9,6	Yes (ADF p= ,034)
Price Shock	2021-2023	26	181,92	112,44	No
Post-Shock	2023-2025	30	89,27	22,7	Yes (ADF p= ,027)

Why is stationarity relevant?

- Stationarity = Constant mean, variance, autocorrelation structure over time
- ARIMA models need stable patterns from the past
- Stationarity was detected using forwards/backwards expanding ADF and KPSS tests

Sparse Data! Most important exogenous Variable (Gas) available only from 2020
Trained on all three Periods (n=108 / n=48), tested on Post Shock (2024+ n=18)

Model Comparion & Evaluation

Training Period	Model	RMSE (Test n=18)
2015-2023 (n=108)	Persistence (B)	43,84
	Seasonal Naive (B)	56,42
	ARIMA (2,1,0)	**28,40**
2020-2023 (n=48)	ARIMA (3,0,0) (B)	26,23
	ARIMAX (3,0,0) + Gas price	**21,56**
	ARIMAX (0,0,0) + Gas price	**21,14** ✓

- Longer training periods bad for prediction (structural breaks)
- Best true ARIMAX model still worse than pure regression (ARIMA is overfitting)
- ARIMAX (0,0,0) / OLS + Gas wins:
  - RMSE
  - Parsimoniousness (OLS 16 obs./parameter v ARIMAX 8 obs./parameter)

Key Findings & Recommendations

Key Findings:

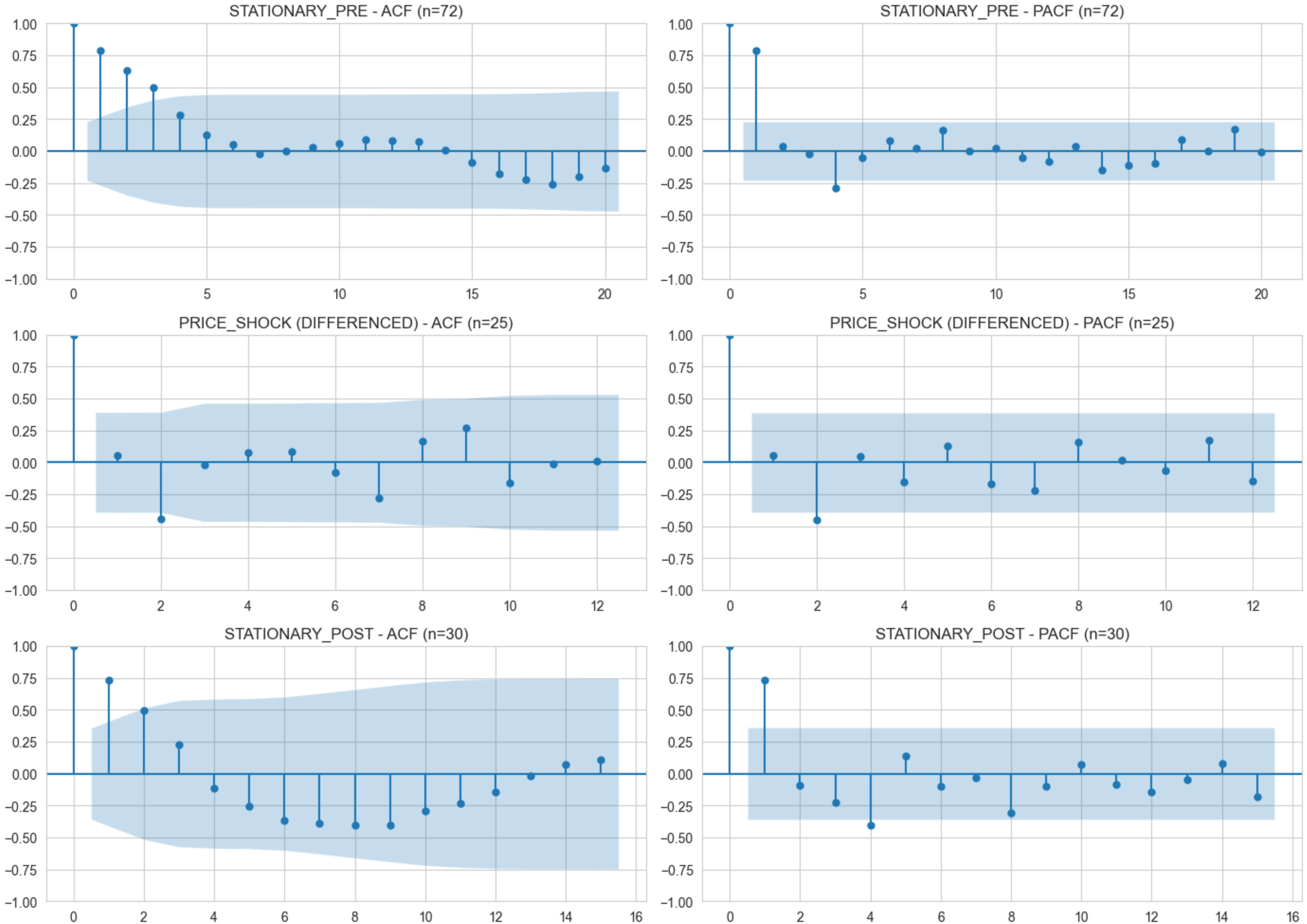
- 1 EUR drop in Gas prices ≈ 1,3 – 1,9 EUR drop in Electricity Price (95% CI)
- Future prices ≈ within +/- 17 EUR/MWh of prediction (95% CI)
- True CIs likely higher, no verifiable trend

Contracting:

- Monitor Gas Prices (OEGPI Indexed 1-month futures) as leading indicator for next 6 months.
- Secure next long term (6 months+) contract at price below 89 EUR/MWh (Current mean).
- Avoid shorter term contracts due to volatility

Test against (3,0,0) model, retrain in 12 month at the latest, log transform:

- Current predictions are too inaccurate for precise forecasts
- Changes in relationship between gas prices and electricity prices likely (regulatory changes, further exogenous shocks)
- Longer training period will increase value of ARIMA(X) model again
- Try log-transformation next time to better deal with variance


STATIONARY_PRE - ACF (n=72)
STATIONARY_PRE - PACF (n=72)
PRICE_SHOCK (DIFFERENCED) - ACF (n=25)
PRICE_SHOCK (DIFFERENCED) - PACF (n=25)
STATIONARY_POST - ACF (n=30)
STATIONARY_POST - PACF (n=30)

Thank you!

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https://github.com/Pringlebot/austrian_electricity_prices_capstone/tree/main